Strategic Product Placement Analysis Using Tableau

1. Project Title

Strategic Product Placement Analysis Using Tableau

2. Project Description

System Overview:

This project analyzes how product placement, pricing, promotions, and consumer behavior impact retail sales.

Using Tableau dashboards, the system transforms raw retail data into interactive visual insights that help

retail managers and analysts make data-driven decisions.

Problem Domain:

Retail analytics and business intelligence where organizations use sales data to optimize product display,

pricing strategies, and promotional campaigns.

Core Technology Used:

• Tableau Public for dashboard creation

• Data visualization techniques

• CSV dataset containing retail sales data

Key Capabilities:

• Product category sales analysis

• Product placement impact analysis

• Pricing comparison with competitors

• Promotion impact on sales

• Consumer segment analysis

Target Users:

• Retail managers

• Business analysts

• Marketing teams

• Students learning data visualization

Real-world Relevance:

Large retail companies use analytics dashboards to optimize product placement strategies, improve sales,

and understand customer purchasing behavior.

3. Application Scenarios

Enterprise Usage:

Retail companies can use dashboards to analyze product placement strategies and increase store sales.

End User Usage:

Store managers can monitor product performance and customer buying patterns.

Educational Usage:

Students can learn practical data visualization techniques using real datasets.

Industrial Usage:

Businesses can use analytics insights to improve marketing strategies, inventory planning, and pricing.

4. Technical Architecture Overview

Frontend Layer:

Interactive dashboards created using Tableau.

Backend Layer:

Tableau data processing engine responsible for filtering, aggregation, and calculations.

Database Layer:

Retail dataset stored in CSV format containing product category, price, promotion status, and sales volume.

Deployment Environment:

Dashboards are published using Tableau Public so they can be accessed via a web browser.

5. Prerequisites

Software Requirements:

- Tableau Public
- Web Browser (Chrome, Edge, Firefox)
- Windows or macOS operating system

Libraries / Frameworks:

No external libraries are required since Tableau includes built-in analytics features.

Hardware Requirements:

- Laptop/Desktop

• Minimum 8GB RAM recommended

• Internet connection for publishing dashboards

6. Prior Knowledge Required

• Basic knowledge of data analysis

• Tableau visualization basics

• Understanding datasets

• Basic business analytics concepts

7. Project Objectives

Technical Objectives:

Create interactive dashboards for analyzing product placement and sales.

Performance Objectives:

Ensure dashboards load efficiently and provide quick insights.

Deployment Objectives:

Publish dashboards on Tableau Public.

Learning Outcomes:

Gain knowledge of business intelligence and data visualization tools.

8. System Workflow

1. Dataset collection

2. Data preprocessing

3. Connecting dataset to Tableau

4. Creating visualizations

5. Building dashboards

6. Generating insights from data

7. Publishing dashboards online

9. Requirement Analysis & System Design

Problem Definition:

Retailers often struggle to determine how product placement affects sales performance.

Functional Requirements:

• Visualize sales by category

• Analyze product placement impact

• Compare competitor prices

• Evaluate promotion effectiveness

Non-Functional Requirements:

• Fast dashboard loading

• User-friendly interface

• Clear visualizations

10. Environment Setup

Steps:

1. Install Tableau Public.

2. Download and prepare dataset.

3. Connect dataset to Tableau.

4. Create worksheets for visualizations.

5. Design dashboards and stories.

11. Core System Development

Feature 1: Category Sales Analysis

Visualization showing average sales volume across product categories.

Feature 2: Product Placement Impact

Analysis of how different display positions influence sales.

Feature 3: Promotion Impact on Sales

Visualization showing how promotional activities affect product sales.

Feature 4: Consumer Segment Analysis

Charts showing how different customer segments contribute to sales.

Feature 5: Top N Products

Identification of top performing products.

12. Integration & Optimization

Component Integration:

All visualizations are combined into dashboards.

Performance Optimization:

Filters and interactive elements are optimized for faster analytics.

Error Handling:

Dataset validation ensures data accuracy.

13. Testing & Validation

Test Cases include dashboard load tests, filter functionality tests, and visualization accuracy tests.

Unit Testing:

Each chart is tested individually.

Integration Testing:

All charts are tested together within the dashboard.

User Acceptance Testing:

Users verify the dashboards meet analytical needs.

Performance Metrics:

Dashboard loading time and interaction responsiveness.

14. Deployment

Deployment Architecture:

Dashboards are deployed on Tableau Public.

Deployment Steps:

1. Finalize dashboards

2. Publish to Tableau Public

3. Share dashboard links with users

15. Project Structure

```
project/
│
├── dataset.csv
├── tableau_dashboard.twbx
├── documentation.docx
```

└── demo_video.mp4

16. Results

The system successfully generated interactive dashboards that provide insights into product placement

strategies, pricing comparison, and promotion impact on retail sales.

17. Advantages & Limitations

Advantages:

• Interactive visualization

• Easy data interpretation

• Supports business decision making

Limitations:

• Dependent on dataset accuracy

• Limited predictive capability

18. Future Enhancements

• Real-time data integration

• AI-based sales prediction

• Mobile dashboard support

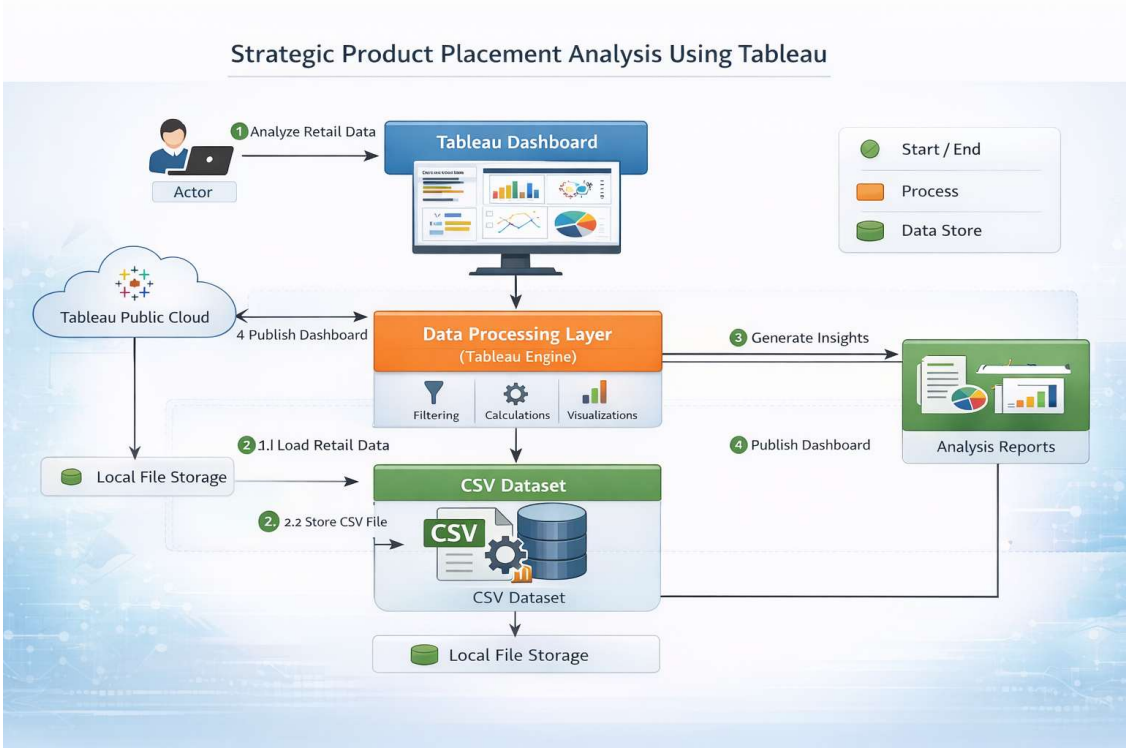
• Automated reporting

19. Conclusion

The project demonstrates how Tableau dashboards can help retailers analyze product placement strategies

and improve sales performance through data-driven insights.

20. Appendix

Includes dataset details, dashboard screenshots, architecture diagrams, DFD diagram, and project files.

System Data Flow Diagram